Master Comparator: Training-Free Few-Shot Gallery Matching for Counterfeit Product Authentication

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***Abstract*—Buyers on e-commerce marketplaces must judge product authenticity from a few listing photographs. We present *Master Comparator*, a training-free, one-class *few-shot* verifier: a test photograph is compared against a small support gallery of k genuine reference images (no counterfeit examples and no learned parameters) by three interpretable checks — ORB keypoint matching, hue–saturation histogram correlation, and OCR brand-text verification — with an optional barcode check, combined by an "allow one miss" vote. On 1,600 photographs of five brands (Nike, Puma, Adidas, Gucci, Louis Vuitton; 93.6% unmodified real photographs) with a strictly held-out gallery of k=20 images (about four per brand), the cascade reaches 58.06% accuracy, 70.48% precision and 27.75% recall (58.5±1.3% accuracy over 20 random galleries), at a mean CPU latency of 155 ms per image. We additionally report: performance as a function of k (cascade F1 rises from 0.03 at k=1 to 0.58 at k=68); comparisons against SIFT, perceptual-hash, HOG and pretrained CNN-embedding baselines on the same gallery, plus a supervised upper reference; robustness under eight capture conditions; and a failure analysis showing that the color check's discriminative power stems largely from background and photography style rather than product color. These results define where a training-free few-shot verifier is and is not suitable for deployment.**

***Index Terms*—counterfeit detection, few-shot verification, training-free, gallery matching, ORB, HSV histogram, OCR, e-commerce, robustness, shortcut learning**

I. Introduction

Counterfeit goods are routinely sold alongside genuine ones on e-commerce marketplaces, and buyers must judge authenticity from a few listing photographs. Subtle indicators — a wrong shade, altered logo proportions, a misspelled brand name — are easy to miss by eye.

Most recent work trains deep classifiers on large, brand-specific corpora [8], [9]. These require labeled counterfeit examples, GPU training and retraining for every new product line. Many practical settings — a small seller, a new product launch, an edge device — have neither counterfeit training data nor a GPU, but *can* supply a handful of photographs of the genuine product.

This paper studies that setting: *one-class, few-shot verification without training*. Given only k genuine reference photographs, can interpretable matching decide whether a new photograph shows a genuine item? Our contributions are:

1) **A training-free few-shot formulation** (Master Comparator) in which logo and color checks score a test image by its best match over a support gallery of k genuine images (Sec. III), with a fully specified, leakage-free evaluation protocol (Sec. V).
2) **A documented five-brand evaluation set** of 1,600 real photographs with a held-out gallery, per-image provenance, and an audit for gallery leakage and duplicated images (Sec. IV).
3) **An extensive empirical study**: a gallery-size (k-shot) sweep with 20 random galleries per size, six baselines plus a supervised reference, robustness under eight capture conditions, and a failure analysis that exposes a background shortcut in the color cue (Sec. VI).

Code, evaluation scripts and an interactive demonstration are public.[1]

II. Related Work

Counterfeit and logo detection. A CNN trained on 3,000 genuine and 2,000 counterfeit sneaker photographs reported over 95% accuracy, outperforming an SVM [8]; the Roboflow "Original or Fake Shoes" dataset provides bounding-box labels for Nike models [7]; fake-logo detectors report over 99% accuracy on curated imagery [9]. All are supervised and need labeled counterfeits.

Few-shot recognition. Matching networks [11] and prototypical networks [12] classify a query by comparing it with a small labeled support set, but learn the embedding by episodic meta-training. Our system shares the support-set protocol but

[1]Code: https://github.com/MalapatiSahasra/fake-product-detection. Demo: https://web-tau-indol-42.vercel.app

TABLE I
POSITIONING AGAINST RELATED WORK

System	Training	Counterfeit labels	Brands	GPU
Footwear CNN/SVM [8]	Yes	Yes	1	Yes
Fake-logo detector [9]	Yes	Yes	Multi (curated)	Yes
Roboflow detector [7]	Yes	Yes	1	Yes
Proto./matching nets [11], [12]	Meta-train	Per episode	–	Yes
This work	**None**	**No**	**5, uncurated**	**No**

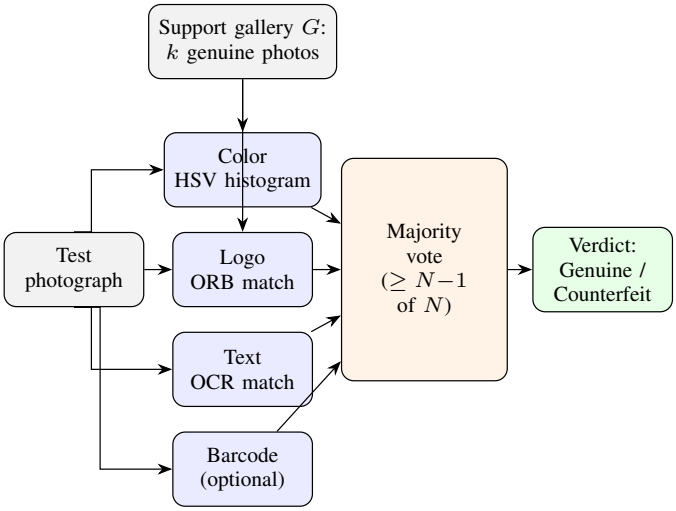


Fig. 1. Master Comparator. Logo and color checks take the best match over the support gallery G; text and barcode checks are gallery-independent. A photograph is accepted if at most one active module disagrees.

learns nothing: comparisons use fixed classical descriptors [1], [2], [10] and need only the genuine class.

Shortcut learning. Classifiers can succeed by exploiting spurious dataset cues such as backgrounds [13]. We test for this explicitly (Sec. VI-G).

Table I summarizes the trade-off: supervised systems report higher accuracy on narrower, curated domains, whereas this work targets zero training cost, no counterfeit labels and CPU-only inference on a heterogeneous pool. Sec. VI-E quantifies the gap to a supervised reference on the same data.

III. PROBLEM FORMULATION AND SYSTEM

A. Why Few-Shot, Not Zero-Shot

Let $G = \{g_1, \ldots, g_k\}$ be a *support gallery* of k photographs known to be genuine. Given a test photograph t, the task is to decide AUTHENTIC or COUNTERFEIT. The setting is *few-shot* because the decision depends on k labeled exemplars of the genuine class (here k=20 across five brands, i.e. about four per brand); it is not zero-shot, which would require recognizing genuineness with no exemplars at all. It is *one-class* because no counterfeit exemplars are used, and *training-free* because no parameter is fitted to data: the gallery is only compared against. Only the OCR check is exemplar-free, as it uses brand strings alone.

B. Gallery Matching and Cascade

Fig. 1 shows the system. The logo and color modules score t by its best match in the gallery:

$$s_{\text{module}}(t) = \max_{g_i \in G} \text{score}(t, g_i) \tag{1}$$

Algorithm 1 Master Comparator Verification

Require: test image t, genuine gallery G, brand strings B, barcode expectation q (optional)
Ensure: verdict $\in \{\text{AUTHENTIC}, \text{COUNTERFEIT}\}$

$r_{\text{logo}} \leftarrow \text{LOGOMATCH}(t, G)$; $r_{\text{color}} \leftarrow \text{COLORMATCH}(t, G)$
$r_{\text{text}} \leftarrow \text{TEXTMATCH}(t, B)$
if barcode check applicable to data source **then**
    $r_{\text{code}} \leftarrow \text{BARCODECHECK}(t, q)$; $N \leftarrow 4$
else
    $N \leftarrow 3$
end if
score $\leftarrow \sum \mathbb{1}[r_m = \text{pass}]$ over active modules
return AUTHENTIC **if** score $\geq N-1$ **else** COUNTERFEIT

Each module m thresholds its score into a pass/fail decision. With N active modules (N=4 when a barcode is expected, otherwise N=3),

$$\text{score}(t) = \sum_{m=1}^{N} \mathbb{1}\left[\text{module}_m(t) = \text{pass}\right] \tag{2}$$

and t is AUTHENTIC iff score$(t) \geq N-1$. All modules always run (Algorithm 1), so every verdict is explainable by its per-module scores.

C. Verification Modules

Logo (ORB). Up to 1,000 ORB keypoints [1] are detected per image; descriptors are matched by brute force under Hamming distance with k-NN (k=2), keeping a match only if it passes Lowe's ratio test [2], $d(m) < 0.75\, d(n)$. The module passes if the best gallery image yields at least τ_{logo}=15 good matches.

Color (HS histogram). Images are converted to HSV and a 50×60 Hue–Saturation histogram is computed (Value excluded for exposure invariance), min–max normalized, and compared by Pearson correlation; the module passes if the best gallery correlation is at least τ_{color}=0.70.

Text (OCR). A bilateral filter and global Otsu threshold precede Tesseract [3] in sparse-text mode (PSM 11). The cleaned text is compared with the brand strings {NIKE, PUMA, ADIDAS, GUCCI, LV} by a sliding-window Ratcliff/Obershelp ratio $2M/T$ [4]; the module passes at τ_{text}=0.80, which tolerates OCR noise but rejects misspellings such as "NUMA" for "PUMA".

Barcode (optional). QR/EAN-13/Code-128 symbols are decoded with ZBar [5] and checked against an expected serial pattern. Only 1 of the 1,565 raw source photographs contains a decodable code, so this module is excluded (N=3) in all experiments; it remains available for authentication-tag photographs.

Complexity. For a gallery of size k, logo matching costs $O(k\, n^2)$ for n keypoints and color matching $O(k\, b)$ for b=3,000 bins; OCR is independent of k. Table II reports measured latency.

TABLE II
SINGLE-THREAD LATENCY PER IMAGE (k=20, 60 IMAGES)

Stage	Mean (ms)	Scales with k?
Decode + resize	3.2	No
Logo (ORB extract + match)	64.3	Yes, $O(k)$
Color (HS histogram)	1.0	Yes, $O(k)$
Text (OCR)	86.1	No
Total	**154.5**	median 136.9, p95 275.1

TABLE III
IMAGES PER BRAND: RAW SOURCE, HELD-OUT GALLERY, EVALUATION SET (GENUINE/COUNTERFEIT)

Brand	Raw (G/C)	Gallery (G)	Evaluation (G/C)
Nike	260 / 253	21	240 / 278
Puma	199 / 91	16	188 / 106
Adidas	178 / 140	14	168 / 160
Gucci	190 / 152	15	178 / 175
LV	28 / 74	2	26 / 81
Total	**855 / 710**	**68**	**800 / 800**

IV. DATASET

A. Source and Composition

Photographs come from the Kaggle "Fake Brand Dataset (Adidas, Nike, Gucci, Puma, LV)" (CC BY-NC 4.0) [6], whose publisher assigns each image a genuine/counterfeit label. The images are uncurated marketplace-style photographs: studio product shots on white backgrounds, handheld photographs, worn-on-foot and lifestyle images. All five brand subsets were used: 855 genuine and 710 counterfeit raw images (1,565). Table III gives the per-brand split at each stage.

B. Held-Out Gallery and Augmentation

A seeded 8% of genuine images per brand (68 images) was reserved as the gallery pool and excluded from evaluation, so no test image can match itself. The remaining 787 genuine and 710 counterfeit images were topped up to exactly 800 per class by deterministic augmentation (seed 42) of a random sample of seeds: rotation in $[-15°, 15°]$, brightness scaling of the V channel in $[0.6, 1.4]$, and Gaussian noise $\sigma \in [5, 20]$. A per-image manifest records the source seed and the exact parameters, so the set is regenerable byte-for-byte.

C. Exact Dataset Statistics

Table IV lists the exact counts, all computed from the per-image manifests. Illumination labels apply only to the 103 augmented images, bucketed by their recorded brightness factor (two decimals); the 1,497 unmodified photographs keep their original lighting. By construction, *multiple images originate from the same photograph* only through augmentation: each augmented image is derived from exactly one recorded source photograph, and 103 source photographs (13 genuine, 90 counterfeit) contribute two images each (original plus one augmented copy). Whether two *different* source photographs show the same physical item is not recorded by the dataset publisher. We therefore measured it with the near-duplicate audit below: 65 evaluation images (4.1%, in 49 same-label pairs) have a near-identical photograph under a different source file and very likely show the same item. Photographs of one item from clearly different angles cannot be detected by this audit, so this figure is a lower bound.

TABLE IV
EXACT EVALUATION-SET STATISTICS (FROM THE PER-IMAGE MANIFESTS)

Statistic	Genuine	Counterfeit	Total
Samples in evaluation set	800	800	1,600
Unmodified photographs	787	710	1,497
Augmented images	13	90	103
Genuine : counterfeit ratio		1 : 1 (800 : 800)	
Held-out gallery pool (not evaluated)	68	–	68
Images per category (brand), evaluation set			
Nike / Puma / Adidas	240 / 188 / 168	278 / 106 / 160	518 / 294 / 328
Gucci / LV	178 / 26	175 / 81	353 / 107
Illumination condition (augmented images, brightness factor b)			
Low light ($b < 0.85$)	4	19	23
Normal ($0.85 \le b \le 1.15$)	7	44	51
Bright ($b > 1.15$)	2	27	29
Original capture (no label)	787	710	1,497
Origin of images			
Distinct source photographs	787	710	1,497
Source photos with 2 outputs	13	90	103
Max. outputs per source photo	2	2	–

TABLE V
FIXED SETTINGS USED IN ALL EXPERIMENTS

Setting	Value
Pre-processing	long side $\le$1,600 px (area interpolation)
ORB / ratio / τ_{logo}	1,000 keypoints / 0.75 / 15 matches
HS bins / τ_{color}	50×60 / 0.70 correlation
OCR / τ_{text}	Tesseract 4.1.1, PSM 11, Otsu / 0.80
Vote	$\ge N-1$ of N=3 (barcode excluded)
Main gallery	k=20 sampled from 68 (seed 42)
Variance study	20 random galleries per k (seeds 1000–1019)
Software	Python 3.10, OpenCV 5.0, scikit-learn 1.7
Hardware	4-vCPU Linux VM, Apple-silicon laptop, no GPU

D. Leakage and Duplicate Audit

Because the source is uncurated, we searched for near-duplicates (pHash Hamming distance ≤ 4 and MobileNetV2 embedding cosine ≥ 0.95) across all 1,668 images. Three genuine test images have a near-duplicate in the 68-image gallery pool (one in the 20-image gallery used below). Eight image pairs appear under *both* labels (16 images; five pairs have identical perceptual hashes), i.e. label noise in the source. A further 49 same-label pairs come from different seed photographs. Removing the 19 leaked or cross-labeled images changes the main result negligibly (F1 0.398 $\rightarrow$ 0.400), so they are retained to keep the set regenerable, and the issue is reported here.

V. EVALUATION PROTOCOL

The protocol is as follows (settings in Table V):

1) **Split.** Gallery pool (68 genuine) and evaluation set (800 genuine, 800 counterfeit) are disjoint by construction (Sec. IV).
2) **Thresholds.** The module thresholds and vote rule are fixed a priori and are *not* tuned on the evaluation set; every image is scored once.

TABLE VI
MAIN RESULT (k=20, 1,600 IMAGES, $\geq$2 OF 3 VOTE)

	Pred. Gen.	Pred. Cf.	Metric	Value
Genuine	TP = 222	FN = 578	Accuracy	58.06%
Counterfeit	FP = 93	TN = 707	Precision	70.48%
			Recall	27.75%
			F1 / FAR	0.398 / 11.6%
20 random galleries (k=20)			Accuracy	58.5 $\pm$ 1.3%
			F1	0.390 $\pm$ 0.046

TABLE VII
CASCADE RESULTS BY BRAND (k=20, SEED 42)

Brand	n	Gallery	Acc.	Prec.	Recall
Adidas	328	7	63.4%	82.4%	36.3%
Gucci	353	3	54.1%	64.3%	20.2%
LV	107	0	69.2%	26.7%	15.4%
Nike	518	6	60.4%	68.8%	26.7%
Puma	294	4	48.6%	74.0%	30.3%

3) **Metrics.** GENUINE is the positive class. We report accuracy, precision, recall, F1, false-acceptance rate FAR $= FP/(FP+TN)$, and threshold-free ROC-AUC of continuous scores.
4) **Gallery variance.** Because results depend on which k images form G, each $k \in \{1, 2, 5, 10, 20, 40, 68\}$ is evaluated on 20 random galleries and reported as mean $\pm$ standard deviation.
5) **Baselines.** Every baseline uses the same gallery and the same best-match rule (Eq. 1). Because baselines have no published thresholds, each continuous score (including our modules') is thresholded by 5-fold cross-validation: the threshold maximizing accuracy on four folds is applied to the fifth. Folds are grouped by source photograph so augmented copies never span folds.
6) **Robustness.** A seeded, class-balanced subset of 400 unaugmented images (200/200) is re-scored under eight perturbations applied to the test image only (the gallery is unchanged).
7) **Latency.** Single-thread wall-clock time over 60 images including decoding (Table II).

VI. EXPERIMENTAL RESULTS

A. Main Result

Table VI reports the cascade on the full evaluation set. The verifier is conservative: 70.5% of accepted photographs are genuine and 11.6% of counterfeits are accepted, but only 27.8% of genuine photographs are recognized. Over 20 random galleries the figures vary by about $\pm$1.3 points of accuracy, so the qualitative profile does not depend on the particular gallery drawn.

B. Per-Brand Results

Table VII also lists how many of the 20 gallery images belong to each brand. Adidas, with the most references (7), has the highest recall and precision; LV has none in this draw and the lowest precision (26.7%, from 26 genuine against 81 counterfeit images). LV's high accuracy simply reflects its 76% counterfeit share. Brand-stratified gallery sampling is therefore recommended in practice.

TABLE VIII
STANDALONE MODULES VS. VOTE RULES (k=20)

Configuration	Acc.	Prec.	Recall	F1
Logo alone	52.3%	52.8%	44.4%	0.482
Color alone	66.9%	69.1%	61.4%	0.650
Text alone	49.9%	45.5%	1.3%	0.024
Cascade, $\geq$ 1 of 3	60.9%	58.0%	79.0%	**0.669**
Cascade, $\geq$ 2 of 3	58.1%	70.5%	27.8%	0.398
Cascade, $\geq$ 3 of 3	50.1%	100.0%	0.3%	0.005

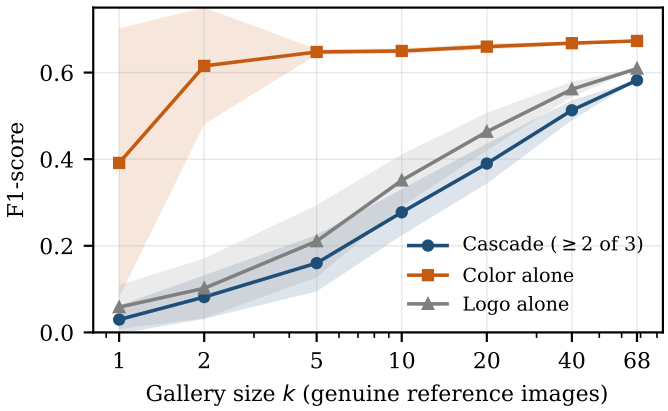


Fig. 2. F1 versus gallery size k (mean $\pm$ s.d. over 20 random galleries; k=68 uses the whole pool).

C. Module Ablation

Table VIII shows that color alone (F1 0.650) outperforms the $\geq$2-of-3 cascade (0.398). OCR read some text in 66% of images, but only 22 of 1,600 contained a legible brand string (10 genuine, 12 counterfeit), so text recall is 1.3% and requiring it to agree discards most of color's correct acceptances. The vote threshold is thus a deployment parameter that trades precision for recall.

D. Effect of Gallery Size (k-Shot)

Fig. 2 shows the few-shot behavior. Cascade F1 grows monotonically from 0.03 (k=1) through 0.28 (k=10) and 0.39 (k=20) to 0.58 (k=68), at which point accuracy is 64.3% and recall 49.7%. Precision decreases slightly with k (73.4% at k=20 to 70.2% at k=68), because a larger gallery also offers counterfeits more chances to match. The color module saturates near F1 0.65 by k=5, while logo matching keeps improving, which shows that recall is limited mainly by gallery coverage.

E. Baseline Comparison

Table IX compares six training-free matchers under the same gallery and best-match rule. Among classical methods, our HS color score achieves the highest AUC (0.705), ahead of perceptual hashing (0.689) and SIFT (0.564). An ImageNet-pretrained MobileNetV2 embedding without fine-tuning is slightly stronger (0.716 at k=20, 0.767 at k=68) but needs

TABLE IX
BASELINES ON THE SAME GALLERY (CROSS-VALIDATED THRESHOLD)

Method (best match over G)	k=20 AUC	k=20 Acc.	k=20 F1	k=68 AUC
Training-free, classical				
ORB matching (ours, logo)	0.517	53.0%	0.382	0.551
HS color (ours)	0.705	68.8%	0.615	0.724
OCR text (ours)	0.389	50.0%	0.292	0.389
SIFT matching [2]	0.564	53.8%	0.549	0.573
Perceptual hash (pHash) [16]	0.689	62.9%	0.582	0.728
HOG cosine [15]	0.346	51.0%	0.137	0.443
Pretrained deep features, no fine-tuning				
MobileNetV2 embedding [14]	0.716	65.8%	0.624	0.767
Uses counterfeit labels (reference only)				
Weighted fusion of our 3 scores (LR)	0.708	66.8%	0.661	0.721
Supervised LR on MobileNetV2	0.854	78.8%	0.790	–
Master Comparator, fixed $\geq$2 of 3	–	58.1%	0.398	–
Master Comparator, fixed $\geq$1 of 3	–	60.9%	0.669	–

TABLE X
ROBUSTNESS (n=400 UNAUGMENTED IMAGES, k=20; PERTURBATION ON TEST IMAGE ONLY)

Condition	Acc.	Prec.	Rec.	FAR	AUC Color	AUC CNN
Clean	56.8	68.5	25.0	11.5	0.724	0.679
Blur (σ=3)	64.7	72.5	47.5	18.0	0.720	0.665
JPEG (q=20)	53.5	56.6	30.0	23.0	0.718	0.680
Low-res ($\frac{1}{4}$ size)	60.0	64.9	43.5	23.5	0.721	0.663
Rotation (30°)	59.3	70.3	32.0	13.5	0.722	0.707
Perspective (15%)	60.8	75.3	32.0	10.5	0.730	0.708
Occlusion (25% area)	48.2	47.7	36.5	**40.0**	0.699	0.562
Low light ($\times$0.45+noise)	50.2	(1/1)	0.5	0.0	**0.339**	0.653
Warm color cast	50.2	60.0	1.5	1.0	**0.219**	0.694

a deep model and no longer gives a per-module explanation. Learning only four weights to fuse our three scores raises F1 to 0.661 without tuning anything else. A supervised logistic-regression probe on MobileNetV2 features, which needs labeled counterfeits, reaches 0.854 AUC and 78.8% accuracy; this marks the price of the training-free, one-class setting on this dataset. Notably, ORB and SIFT keypoint matching are near chance (AUC 0.52–0.57) across brands, as expected for a gallery spanning many different products.

F. Robustness under Practical Capture Conditions

Table X tests eight conditions typical of buyer- or seller-captured photographs. Geometric changes (rotation, perspective) are tolerated. Blur, heavy JPEG compression and low resolution *raise* recall but also raise the false-acceptance rate (to 18–24%), because smoothing produces generic keypoints that match more gallery images. A 25%-area occlusion is the most harmful case for security, accepting 40% of counterfeits. Low light and a warm color cast make the color module fail completely: its AUC falls *below* chance (0.34 and 0.22) and the cascade rejects almost everything, whereas the CNN embedding stays near 0.65–0.69. Photometric normalization (gray-world white balance with chromatic masking) did not recover the color cue (AUC 0.46 under low light, 0.41 under color cast). The system should therefore be deployed only with controlled capture (consistent lighting and white balance, unobstructed view).

TABLE XI
ERROR ANALYSIS BY NEAR-WHITE BACKGROUND FRACTION

Background fraction	n	Genuine	Recall	FAR
$<$0.10	426	130	6.9%	4.4%
0.10–0.30	392	124	5.6%	8.2%
0.30–0.60	254	103	31.1%	19.9%
$\geq$0.60	528	443	39.3%	32.9%

G. Failure-Case Analysis

Fig. 3 shows typical errors, and they follow a clear pattern. Genuine photographs taken by hand or in lifestyle settings are rejected, while counterfeit studio shots on white backgrounds are accepted. To quantify this we measured each image's fraction of near-white pixels (S$<$30, V$>$200). Genuine evaluation images average 52% background against 24% for counterfeits, and the gallery averages 50%. Background fraction *alone* separates the classes with AUC 0.747, higher than the color module (0.705). When the color histogram is restricted to chromatic product pixels (S$\geq$40, V$\geq$40), its AUC drops to 0.466, i.e. chance. The color module's apparent discriminative power therefore comes mainly from *photography style*, a dataset shortcut [13], rather than from product color. Table XI confirms this: recall is 39.3% on white-background images but under 7% on others, and false acceptances concentrate on white-background counterfeits (FAR 32.9%).

Module-level patterns support this reading. Of the 578 false negatives, 272 passed only color (failed logo), 136 passed only logo, and 168 failed all modules. Of the 93 false positives, 86 passed both logo and color, i.e. counterfeit studio shots that resemble the studio-style gallery. Some cross-labeled duplicates (Sec. IV-D) are also counted as errors although they are visually indistinguishable.

VII. DISCUSSION

What the results support. With about four genuine references per brand and no training, the verifier is conservative (precision 70.5%, FAR 11.6%) and fully explainable at 155 ms per image on a CPU. Recall rises steadily with gallery size (Fig. 2), and the vote threshold or a small learned weighting moves the operating point (Tables VIII, IX).

What they do not support. The results do not support unsupervised brand-level authentication of arbitrary marketplace photos. On this dataset the color cue largely detects photography style, keypoint matching across different products is near chance, and occlusion or lighting changes either raise false acceptance (40% under occlusion) or disable the color check. We therefore scope the deployment claim: Master Comparator is suited to *controlled-capture, per-SKU verification* — for example a checkpoint or seller-onboarding flow in which the item is photographed against a fixed background under fixed lighting and compared with references of the same SKU — and as a first-stage triage that routes uncertain listings to



Fig. 3. Representative errors (k=20). (a) Genuine Adidas, handheld on carpet: best color correlation 0.18, 16 ORB matches → rejected. (b) Genuine Gucci, worn outdoors: color 0.19, 10 matches → rejected. (c) Counterfeit Gucci and (d) counterfeit Nike studio shots on white: color 1.00/1.00, 15/17 matches → accepted.

human review. It should not be used as the sole authority on unconstrained marketplace images.

Barcode. 1,564 of 1,565 raw photographs carry no readable code, so barcode verification needs deliberately captured authentication tags.

A. Threats to Validity

Labels and duplicates. Labels come from the dataset publisher; eight cross-labeled duplicate pairs show they are imperfect (Sec. IV-D). **Independence.** Whether different raw photographs depict the same physical unit is undocumented; 49 same-label near-duplicate pairs suggest some repetition. **Single source.** All images come from one Kaggle dataset, and its background bias may not transfer to other sources. **Synthetic perturbations.** The robustness conditions approximate, but do not reproduce, real capture variation. **Gallery sampling.** Mitigated by reporting 20 random galleries per k.

VIII. Conclusion and Future Work

We presented Master Comparator, a training-free, one-class few-shot verifier that compares a product photograph with a small gallery of genuine references using ORB, hue–saturation and OCR checks combined by an "allow one miss" vote. On 1,600 real five-brand photographs with k=20 references it reaches 58.06% accuracy, 70.48% precision and 27.75% recall at 155 ms per image on a CPU, and F1 rises from 0.03 to 0.58 as k grows from 1 to 68. Compared on the same gallery, its color cue is the strongest classical matcher, close to a pretrained CNN embedding, while a supervised probe shows the remaining gap. Robustness and failure analyses show that the color cue depends on photography style and lighting and that occlusion raises false acceptance, which restricts deployment to controlled-capture, per-SKU verification. Future work: (1) background segmentation before color matching, (2) brand-stratified and per-SKU galleries, (3) learned weighting of module scores, (4) a hybrid with pretrained embeddings for gallery ranking, and (5) evaluation on authentication-tag and multi-source imagery.

Acknowledgment

Evaluation photographs are from the "Fake Brand Dataset (Adidas, Nike, Gucci, Puma, LV)" published on Kaggle by prashanthgolla under CC BY-NC 4.0 [6].

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